Harsh Vaibhav

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Summary

2026 B.Tech graduate from NIT Jalandhar and a software engineer working across Python and the MERN stack. I have shipped production FastAPI services with async processing and background job orchestration, built multi-agent LangGraph workflows with RAG pipelines, and delivered full stack applications end to end with React, Node.js, Express, and MongoDB.

Education

National Institute of Technology Jalandhar — India
Bachelor of Technology | CGPA: 7.58/10 — *2022 – 2026*

Experience

AI Automation Engineer — Feb 2026 – Apr 2026
ERP BOSS — *Remote*

- Architected an 8-node multi-agent LangGraph workflow for email triage, classification, and response generation, integrating a RAG pipeline with Groq LLMs and Gemini embeddings to deliver context-grounded replies across inboxes with 50+ unread threads
- Evaluated inference-layer tradeoffs across latency, reasoning depth, and cost, selecting Groq for low-latency responses and Gemini for reasoning-heavy drafts to guide production model selection.
- Built Gmail API integration using OAuth 2.0 for automated retrieval, threading, and dispatch, reducing end-to-end email handling workload by 80% with zero manual intervention.
- Implemented a RAG-based knowledge retrieval layer with document chunking, embeddings, and vector search, enabling context-aware responses and improving the reliability of automated email generation.

Full Stack Developer Intern — Dec 2025 – Feb 2026
DeepQ AI — *In-office*

- Developed and maintained 15+ RESTful APIs using Django REST Framework to support core productivity features including user authentication, employee management, and workflow automation.
- Architected 5+ PostgreSQL database schemas with optimized queries for efficient storage and retrieval of employee records, workflow data, and real-time activity logs spanning 10+ database tables.
- Delivered 10+ responsive React.js interfaces including dashboards, reporting modules, and notification systems, collaborated in a 4-member cross-functional team to deliver production-ready features bi-weekly.
- Implemented JWT-based authentication and role-based access control across backend services, securing 10+ API endpoints and integrating frontend workflows with RESTful APIs for seamless data synchronization.

SDE Intern — June 2025 – Aug 2025
Basil Infotech — *Remote*

- Delivered a full-stack School Management System (EduNexa) using React, Node.js, Express.js, and MongoDB.
- Architected and deployed RESTful APIs with query indexing and join optimization, cutting average response latency.
- Secured all routes with role-based access control (RBAC) and JWT-based session management, eliminating unauthorized access incidents; applied Redis caching and pagination, reducing server load by 20% under concurrent traffic.
- Implemented API testing with Postman and resolved integration issues, improving application reliability across core workflows.

Projects

OpsPilot | *React, FastAPI, Python, ChromaDB, Groq, Docker* | Live — July 2026

- Shipped a full-stack Retrieval-Augmented Generation (RAG) platform with a React frontend and Python FastAPI backend, featuring asynchronous background ingestion and real-time job-status tracking, deployed live at sub-3-second p50 query latency.
- Built hybrid semantic search on ChromaDB (dense embeddings + BM25 via reciprocal rank fusion) over a streaming chunking pipeline handling 80+ page PDFs at constant memory, lifting recall@5 from 0.74 to 0.84, and from 0.53 to 0.77 on exact-ID lookups on a 50-query gold set.
- Extended the platform into a conversational AI channel with streaming speech-to-text, custom silence-based endpointing, a dedicated LLM call for speech-optimized responses, and neural text-to-speech with multi-provider fallback.
- Source Code

Skills

Languages: Python, C++, JavaScript, SQL
Agentic AI & LLMs: LangGraph, LangChain, multi-agent orchestration, RAG pipelines, prompt engineering, LLM integration (Groq, Gemini), RAGAS evaluation
Backend & APIs: FastAPI, Node.js, Express.js, REST API design, Pydantic, OAuth 2.0, JWT
Databases & Data: PostgreSQL, MongoDB, SQLite, Redis, ChromaDB, Pinecone, indexing, Pandas, NumPy
Cloud & DevOps: Docker, AWS, Git/GitHub, Linux
Frontend & CS Core: React.js, DSA, Operating Systems, DBMS, Computer Networks, System Design

Achievements

Smart India Hackathon (SIH) 2025 — National-level qualifier, 100,000+ participants, representing NIT-J with an AI-powered solution
Solved 850+ problems over various coding platforms like LeetCode, Codeforces and GeeksforGeeks etc.